2022

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-0902

[Image Processing (Elective-I)]

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1, 2** and ***any four*** questions from the rest.

1. Answer ***any five*** questions from the following : 2×5

(a) How does the Weber Ratio affect the brightness of an image?

(b) Consider the continuous function $f(t) = \sin(2pnt)$. What would the sampled function look like, if $f(t)$ is sampled at the Nyquist rate with samples taken at $t = 0, \Delta T, 2\,\Delta T, ...$?

(c) Perform shear (vertical) and shear (horizontal) transformations on the following matrix :

1	3	1	3
2	4	6	1
2	6	5	3
5	3	4	2

(d) How can median filters be used in the noise reduction of an image?

(e) What is the impact of gamma in gamma transformation used in image enhancement?

(f) What are the principal types of data redundancy that occur in images?

(g) How does correlation differ from convolution? Give an example.

2. Answer ***any five*** questions from the following : 4×5

(a) 'Histogram matching' is a useful contrast manipulation technique that transforms an image's histogram to match the one of another image. Describe clearly how you can achieve it.

(b) The following figure shows :

(i) A 3-bit image of size 5×5 image in the square, with x and y coordinates spcified,

(ii) A Laplacian filter,

(iii) A low-pass filter.

Y/X	Image				
	1	2	3	4	5
1	3	7	6	2	0
2	2	4	6	1	1
3	4	7	2	5	4
4	3	0	6	2	1
5	5	7	5	1	2

Laplacian mask		
0	1	0
1	−4	1
0	1	0

Low pass filter		
0.01	0.1	0.01
0.1	0.56	0.1
0.01	0.1	0.01

Compute the following :

(I) The output of a 3×3 mean (average) filter at (3, 3).

(II) The output of the 3×3 Laplacian filter shown above at (3, 3).

(III) The output of the 3×3 low-pass filter shown above at (3, 3).

(IV) The histogram of the whole image.

(c) Cluster-seeking algorithms can be applied to unsupervised learning. Justify.

(d) What is the motivation of sharpening filters? Critically comment on the following statement : "Second derivative for filter enhances fine detail much better than the first derivative".

(e) Show all the bit planes of the following image :

7	6	5
4	3	2
1	1	0

(f) If an image looks too dark or bleached out, then which transformation is used to make it more prominent? How will the appropriate correction be used?

(g) Explain the most commonly used distance measures used in image processing.

3. (a) Plot the histogram of the following 8×8 image :

0	5	7	7	5	8	7	8
7	2	6	2	6	5	6	8
6	9	7	7	0	7	2	7
6	6	1	7	6	7	7	5
9	6	0	7	8	2	6	7
2	8	8	2	7	6	7	8
7	3	2	6	1	7	5	8
9	9	5	6	7	7	7	7

(b) Perform the histogram equalization of the image. Output the resultant image and its corresponding histogram.

(c) Write the application of sharpening filters. Consider the following image segment :

15	6	21	22
17	15	6	19
14	3	11	12
19	14	19	16

Perform the following transformation on the shaded pixels : Log in [0, 8]. 3+4+3

4. (a) Find out the Huffman encoding and the code length of a message containing the following characters and frequency. Show the steps of source reduction and draw the Huffman tree.

Character	Frequency
a	2
b	8
c	9
d	17
e	25
f	50

(b) What is Pseudocolour image processing? What is the principal use of pseudocolour image processing? Explain the different methods to perform pseudocolour image processing . 5+5

5. (a) Perform zooming and shrinking of the following image to double the size using (i) bilinear and (ii) bicubic interpolation.

1	4	5	2
3	4	6	5
2	3	4	1
3	3	1	2

(b) Define spatial resolution and intensity resolution. How can you choose the spatial resolution? Jusitfy your answer. 6+4

6. (a) Convert the following 3-bit RGB image to CMY model :

1,2,5	4,1,5	5,4,2	2,1,4
3,6,5	4,1,3	6,6,1	5,4,1
2,1,4	3,1,4	4,2,3	1,4,3
3,2,1	3,1,3	1,1,2	2,3,1

(b) Explain the different noise models. Mention its applications in different types of noise. 6+4

7. (a) Discuss the process of region splitting and merging for region-based segmentation.

(b) Apply gray-level slicing on the given image, where $r_1 = 3$ and $r_2 = 6$ (i) with background and (ii) without background.

3	4	5
6	3	7
1	2	4

4+6

8. (a) Cluster the dataset $\{(0, 0)', (0, 1)', (5, 4)', (5, 5)', (4, 5)', (1, 0)'\}$ by

(i) applying the Maximin Distance Algorithm and

(ii) the k-means algorithm.

(b) How can gradient operators be used for Edge Detection? 8+2